Mark schemes

Q1.

(a) $14a = 14g \sin 45° - T$

Equation of motion for one particle

M1

$6a = T - 6g$

Correct equation

A1

$14a = 14g \sin 45° - (6a + 6g)$

Equation of motion for other particle

Correct equation

M1 A1

$a = \dfrac{14g \sin 45° - 6g}{20} = 1.91 \text{ ms}^{-2}$

Correct a from correct working

M1 A1

6

(b) $T = mg$

Equation for one particle

M1

$T = 14g \cos 45°$

Equation of other particle

correct m

M1 A1

$m = 14 \cos 45° = 9.90$ kg

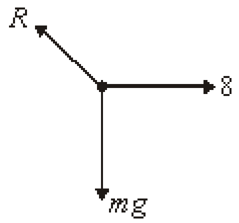
A1

4

[10]

Q2.

(a)



Correct force diagram with labels

B1

1

(b) $R \cos 30° = 8$

Resolving horizontally to get two terms

M1

Correct equation

A1

$R = \frac{8}{\cos 30°} = 9.24$

***ag** Correct answer from correct working (other methods: M1 A1 if correct)*

A1

3

(c) $R \cos 60 = 9.8m$

Resolving horizontally to get two terms, with 8 not included

M1

Correct equation

A1

$m = \frac{8 \cos 60°}{9.8 \cos 30°} = 0.47$

Correct m to 2 sf

A1

3

(Resolving perpendicular to the plane: M1A1 for equation and A1 for final answer)

[7]

Q3.

(a) $R = 20 \times 9.8 = 196$

cao

B1

1

(b) $F \le 0.3 \times 196 = 58.8$

Using 0.3 × 196

M1

If $P = 80$, $F = 58.8$

58.8 as answer

A1

If $P = 40$, $F = 40$

40 as answer

A1

3

(c) $P - 58.8 = 20 \times 0.8$

Three term equation of motion including 58.8

M1

Correct equation

A1

$P = 74.8$

A1

Correct P

A1

3

(d) $a = \dfrac{-58.8}{20} = -2.94$

Use of F = ma with ± 58.8

M1

Correct acceleration with a negative sign

A1

$0^2 = 6^2 + 2 \times (-2.94)s$

Use of $v^2 = u^2 + 2as$ with $v = 0$

m1 A1

$s = \frac{36}{5.88} = 6.12$

Correct distance

A1

5

[12]

Q4.

(a) $T \cos 42° = 60 \cos 40°$

Resolving horizontally

M1

Correct equation

A1

$T = \frac{60\cos 40°}{\cos 42°} = 61.8$ N (to 3sf)

awrt 61.8

A1

3

(b) $T \sin 42° + 60 \sin 40° = 9.8m$

Resolving vertically with three terms

M1

Correct equatino with T from bove

A1

$m = \frac{T\sin 42° + 60\sin 40°}{9.8} = 8.16$ kg (to 3sf)

Solving for m

m1

awrt 8.16

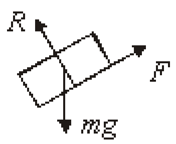
A1

4

[7]

Q5.

(a)



Correct force diagram

B1 1

(b) $R = 7 \times 9.8\cos30° = 59.4$ N

Resolving perpendicular to plane

Correct force

M1/A1 2

(c) $F = 7 \times 9.8\sin30° = 34.3$ N

Resolving parallel to plane

Correct force

M1/A1 2

(d) $34.3 \le \mu \times 59.4$

Using F = μR *or* F $\le \mu$R

M1

$\mu \ge 0.577$

Correct inequality

A1 2

(e) Remains at rest, as mass appears in both F and R, so inequality unchanged.

Conclusion

B1

Reason

B1 2

[9]